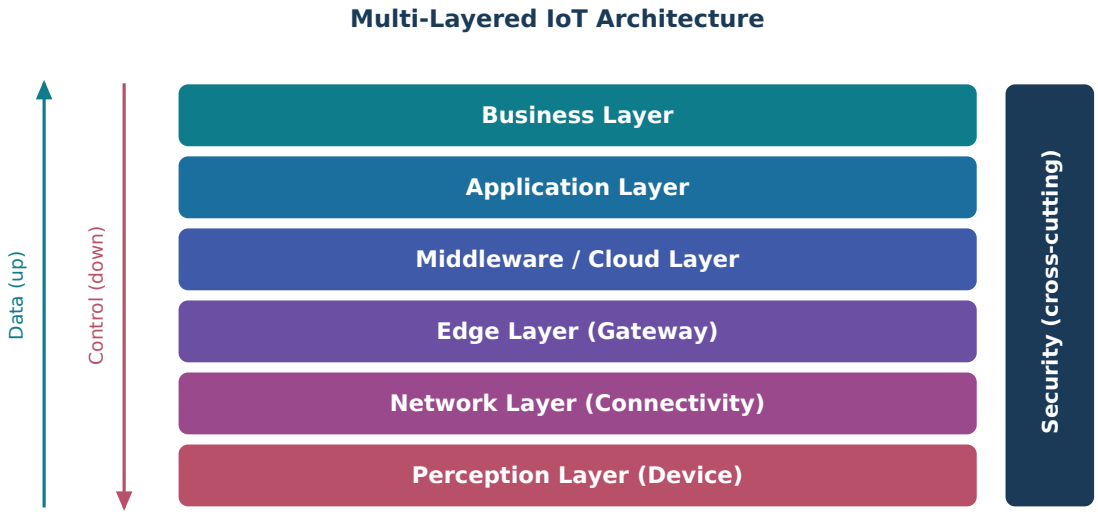
Unit 2: Principles and Architecture of IoT Systems

Question Bank & Answer Key

Section A: 10-Mark Questions

Q1. Explain the multi-layered IoT architectural framework in detail. Discuss the core functions, components, and interactions associated with each layer from the physical device up to the business layer.

Answer:



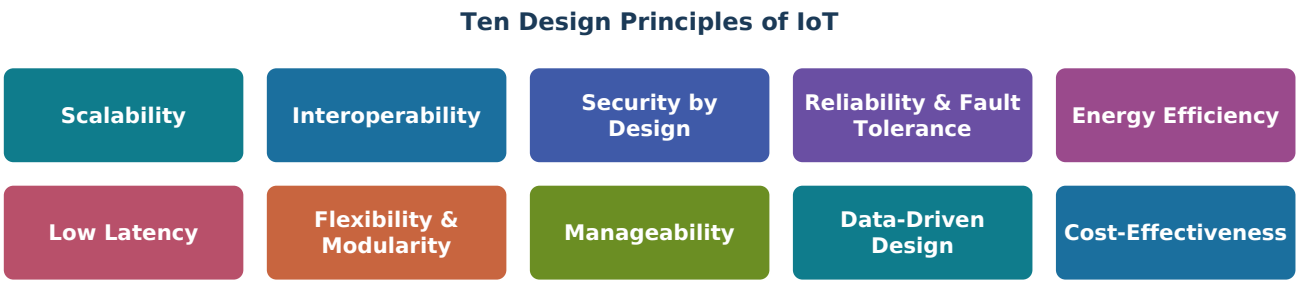
1. **Perception Layer (Device Layer):** The bottom-most physical layer responsible for interacting directly with the physical environment to capture data and perform real-world actuation. This layer consists of sensors (e.g., temperature, soil moisture, motion) that collect environmental attributes and actuators (e.g., motors, water pumps) that execute control commands. It handles physical hardware parameters including battery life constraints, memory, and ruggedization based on deployment conditions.
2. **Network Layer (Connectivity Layer):** The communication backbone that securely transports data collected by perception devices to processing platforms. It integrates short-range wireless protocols (Bluetooth, ZigBee, Wi-Fi) and long-range LPWAN or cellular links (LoRaWAN, NB-IoT, 4G/5G). Standard messaging protocols such as MQTT, CoAP, and HTTP operate at this layer to guarantee efficient payload delivery across heterogeneous networks.
3. **Edge Layer (Gateway Layer):** An intermediate physical or virtual layer located between local end devices and cloud infrastructure to process data near the source. Edge gateways (e.g., Raspberry Pi or industrial edge nodes) perform local data aggregation, filtering, noise reduction, and preliminary security checks. By handling time-critical logic locally, the edge layer minimizes latency and drastically reduces cloud bandwidth usage.
4. **Middleware / Cloud Layer:** The centralized core platform responsible for device management, data routing, storage, and analytics execution. Cloud middleware (e.g., AWS IoT Core, Azure IoT Hub) provides core services such as device registration, authentication, and SQL/NoSQL time-series data storage. It runs real-time streaming pipelines, batch analytics,

and handles event triggering across scalable cloud resources.

5. **Application Layer:** The user-facing software interface that translates raw IoT data into visual insights and actionable controls. This layer includes mobile applications, web dashboards, and control panels equipped with data charts, maps, and automated alerts. It allows operators to monitor system health remotely and send commands back down to physical actuators.
6. **Business Layer:** The high-level management layer that aligns IoT operational data with business logic, monetisation models, and enterprise governance. It handles commercial analytics, return on investment (ROI) calculations, automated billing models (e.g., usage-based water metering), and strategic decision-making. It integrates IoT capabilities into broader enterprise platforms like CRM and ERP systems.
7. **Security Layer (Cross-Cutting):** An overarching framework providing end-to-end data privacy, device integrity, and threat defense across all layers. It enforces secure device identities, TLS/DTLS encryption for data in transit, secure boot, and digitally signed over-the-air (OTA) updates. Security must span continuously from the physical sensor up to cloud applications.
8. **Scalability Factor in Architecture:** The architectural capability to expand seamlessly from small deployments to massive device fleets. Multi-layered architectures decouple individual tiers, ensuring message brokers, database systems, and networks can handle horizontal growth from tens to millions of connected devices without structural redesign.
9. **Interoperability Factor in Architecture:** The structural provision allowing heterogeneous devices from diverse manufacturers to communicate effectively. Standardized data formats (JSON, CBOR, XML) and open communication standards (MQTT, CoAP, OPC UA) are implemented across layers to eliminate proprietary lock-in.
10. **Energy Efficiency and Resource Constraints:** Optimization strategies engineered into the architecture to prolong operational lifetimes of battery-powered nodes. The architecture incorporates lightweight communication protocols, duty-cycling schemes, and adaptive data transmission rates to conserve power at the perception and network layers.

Q2. Elaborate on the ten core Design Principles of IoT Systems. Provide clear technical explanations and practical industry examples for each principle.

Answer:

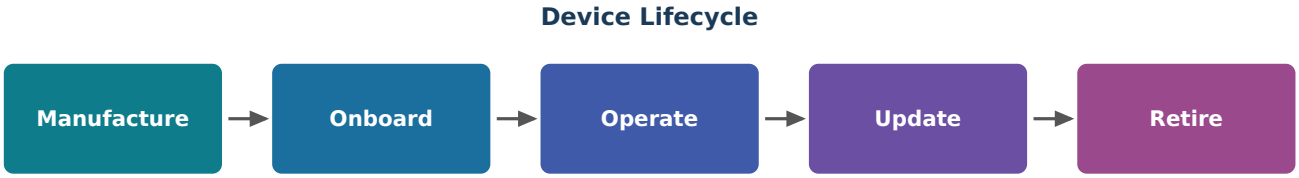


1. **Scalability:** The architectural ability of an IoT system to accommodate continuous growth in connected devices and data traffic without performance degradation. *Example: Smart city lighting platforms engineered to scale seamlessly from 100 localized streetlights to millions across an entire metropolis without redesigning cloud brokers.*
2. **Interoperability:** The seamless integration and exchange of data across heterogeneous devices, software platforms, and protocols from multiple vendors. *Example: Combining environmental sensors from different manufacturers using unified data payloads like JSON over standard MQTT message brokers.*

3. **Security by Design:** Embedding security controls into hardware, firmware, network protocols, and software layers from the initial design phase rather than as an afterthought. *Example: Equipping every hardware node with cryptographically unique device IDs, Secure Boot validation, and digitally signed OTA firmware updates.*
4. **Reliability and Fault Tolerance:** Ensuring uninterrupted system functionality and data integrity despite network outages, hardware glitches, or power drops. *Example: Utilizing store-and-forward edge buffering so agriculture moisture sensors store data locally during cellular drops and transmit when reconnected.*
5. **Energy Efficiency:** Minimizing power consumption through low-power hardware design, duty cycling, and lightweight protocols to maximize battery life. *Example: Remote asset tracking sensors sleeping 99% of the time and waking up briefly using low-power BLE or NB-IoT bursts to extend battery life past 10 years.*
6. **Low Latency and Responsiveness:** Minimizing response times between signal detection and actuation in time-critical operational environments. *Example: Deploying local edge computing nodes on smart factory floors to execute immediate emergency shutdowns within milliseconds without waiting for cloud round-trips.*
7. **Flexibility and Modularity:** Structuring systems into decoupled, independent modules so protocols, databases, or hardware components can be swapped easily. *Example: Switching cloud database middleware from SQL to time-series databases without modifying edge gateway firmware or perception protocols.*
8. **Manageability:** Providing centralized remote control capabilities for provisioning, health monitoring, software upgrading, and retiring devices. *Example: Maintaining a centralized device registry that tracks fleet battery levels, signal health, and pushes mass remote OTA firmware updates.*
9. **Data-Driven Design:** Structuring data ingestion pipelines into hot, warm, and cold paths based on analytics urgency and control requirements. *Example: Directing real-time safety telemetry to hot edge processing paths while sending historical usage logs to cold long-term cloud storage.*
10. **Cost-Effectiveness:** Balancing upfront hardware expenses, operational cloud costs, bandwidth charges, and maintenance expenditures to ensure strong ROI. *Example: Utilizing compressed binary data formats like CBOR over UDP to cut cellular data transmission bills for remote smart meters.*

Q3. Examine the core Needed Capabilities in modern IoT systems across device management, communication, data processing, security, and edge operations.

Answer:

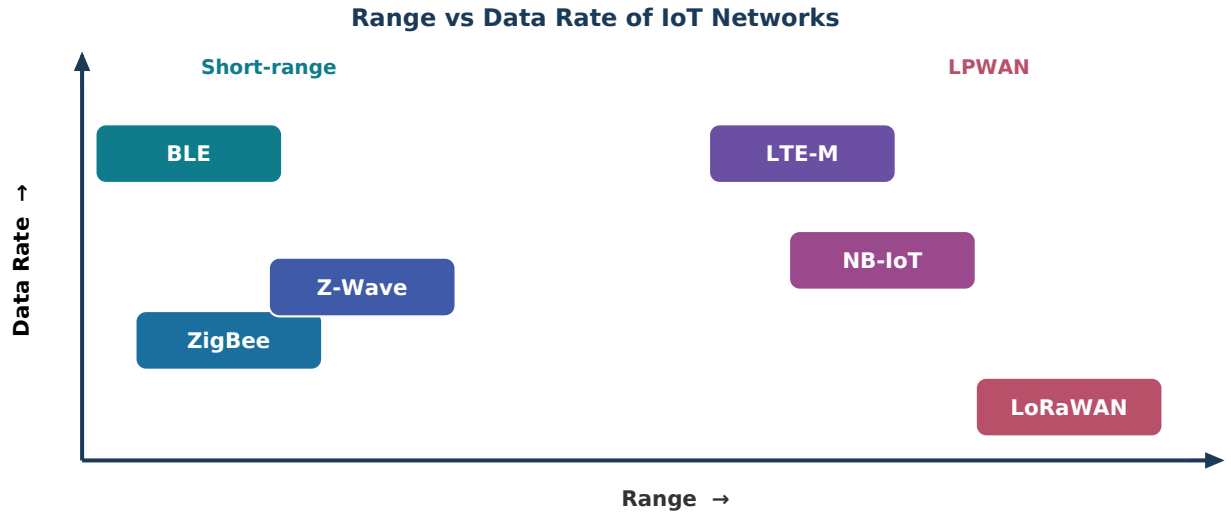


1. **Device Management & Provisioning:** Features enabling automated zero-touch onboarding, secure identity assignment, configuration management, and lifecycle tracking of devices. Assigns unique cryptographic keys, manages fleet health metrics (battery, signal), and executes remote Over-The-Air (OTA) firmware updates.
2. **Heterogeneous Communication & Networking Support:** Native compatibility with multiple short-range, long-range, cellular, and mesh wireless protocol standards. Implements message translation and Quality of Service (QoS) guarantees (e.g., exactly-once, at-least-once) across protocols like MQTT, CoAP, BLE, and LoRaWAN.
3. **Data Management & Ingestion Pipelines:** High-throughput data ingestion pipelines supporting real-time streaming analytics and long-term batch analytics. Handles continuous stream processing, data compaction, lifecycle retention policies, and historical storage in specialized time-series databases.
4. **Security, Privacy & Compliance:** Comprehensive protection mechanisms covering authentication, role-based authorization, data encryption, and regulatory compliance. Utilizes TLS/DTLS protocols, identity access management (IAM), anomaly detection for compromised nodes, and compliance with standards like GDPR, HIPAA, and IEC 62443.
5. **Edge Computing & Local Processing:** Decentralized computing capabilities executed directly on edge gateways or local devices near the data source. Performs local data filtering, runs lightweight AI/ML inference models (e.g., predictive maintenance), and enables offline functionality during network outages.
6. **Integration & API Management:** Standardized programmatic interfaces enabling external enterprise software and applications to interact with the IoT backend. Exposes RESTful and GraphQL APIs for web applications, along with event-driven WebSockets for real-time integration with ERP, CRM, and SCADA platforms.
7. **Observability & Operations Monitoring:** Real-time monitoring tools providing visibility into overall fleet status, network health, and system bottlenecks. Captures distributed traces, metrics, and logs from nodes to drive operational dashboards (Grafana/Kibana) and automated self-healing retry triggers.
8. **Scalability & Multi-Tenancy:** System design provisions allowing shared platform deployment across isolated customer environments and expanding workloads. Supports multi-tenant isolation, horizontal auto-scaling of message brokers, and elastic storage resource expansion under spike loads.
9. **User Interface & Application Delivery:** High-level visualization platforms designed for human operators and end-users. Delivers web portals, mobile apps, spatial maps, charts, predictive insight reports, and remote actuator trigger controls.
10. **Secure Lifecycle Decommissioning:** Managed operational processes covering every stage of hardware deployment from initial manufacturing to retirement. Manages the full lifecycle path (Manufacture → Onboard → Operate → Update → Retire), revoking security certificates

safely upon device decommissioning.

Q4. Provide a comprehensive classification and comparative analysis of IoT Communication and Networking Standards, detailing short-range, LPWAN, and IP-based routing technologies.

Answer:



1. **Bluetooth Low Energy (BLE):** A short-range, ultra-low-power wireless protocol optimized for periodic, small data payload bursts. Operates in 2.4 GHz band, range 10–30 meters, data rate up to 1 Mbps; supports point-to-point, broadcast, and mesh topologies.
2. **ZigBee Protocol:** A low-power, short-range wireless standard based on IEEE 802.15.4 designed for home and industrial automation. Utilizes mesh topology to relay signals across multiple nodes, offering high reliability and low power consumption over short ranges.
3. **Z-Wave Technology:** A proprietary, sub-GHz short-range mesh networking protocol designed specifically for smart home automation. Operates in sub-GHz ISM bands (868/908 MHz) to avoid Wi-Fi interference; supports AES-128 encryption and ranges up to 100 meters.
4. **LoRaWAN (Long Range Wide Area Network):** An open LPWAN protocol operating on unlicensed sub-GHz spectrum, managed by the LoRa Alliance. Offers extreme range (10–15 km rural), 5–10+ year battery life, low data rates (0.3–50 kbps), and a star-of-stars network topology.
5. **NB-IoT (Narrowband IoT):** A cellular LPWAN standard developed by 3GPP operating in licensed spectrum within LTE bands. Delivers deep indoor signal penetration, low power life (10+ years), fixed static sensor support, and data rates up to 250 kbps.
6. **LTE-M (LTE Cat-M1):** A 3GPP cellular LPWAN protocol engineered for IoT applications requiring higher data speeds, mobility, and voice support. Provides data rates up to 1 Mbps, low latency (~50–100 ms), cellular roaming handover, and support for Voice-over-LTE (VoLTE).
7. **IETF 6LoWPAN:** An adaptation layer protocol that enables IPv6 packets to transmit over low-power IEEE 802.15.4 wireless networks. Compresses heavy IPv6 headers and handles packet fragmentation, giving tiny microcontrollers native IP addresses.
8. **RPL (Routing Protocol for Low-Power and Lossy Networks):** An IPv6-based distance-vector routing protocol designed specifically for lossy, constrained IoT networks. Builds Destination-Oriented Directed Acyclic Graphs (DODAGs) optimized for multipoint-to-point data collection to central sinks.
9. **Network Selection Criteria - Bandwidth vs. Range:** Trade-offs inherent in choosing wireless IoT protocols based on operational requirements. Short-range protocols (BLE/Wi-Fi)

trade coverage for higher bandwidth, whereas LPWAN (LoRaWAN/NB-IoT) sacrifices data rate to achieve long range and multi-year battery life.

10. **Licensed vs. Unlicensed Spectrum Standards:** Regulatory distinction governing LPWAN deployment models. Unlicensed LPWAN (LoRaWAN) allows private network setups without subscription costs, while licensed LPWAN (NB-IoT/LTE-M) relies on telecom operator coverage and guaranteed Quality of Service.

Q5. Analyze the core Messaging, Data Representation, and Device Management Standards in IoT. Detail the operation of MQTT, CoAP, AMQP, CBOR, and OMA LwM2M.

Answer:



1. **MQTT (Message Queuing Telemetry Transport):** An extremely lightweight, publish-subscribe messaging protocol operating over TCP/IP. Relies on a central Message Broker that routes messages from publishing devices to subscribing clients based on topic strings, minimizing message header overhead.
2. **CoAP (Constrained Application Protocol):** A lightweight web transfer protocol designed by IETF for resource-constrained nodes, mirroring REST architecture over UDP. Maps directly to HTTP methods (GET, POST, PUT, DELETE) but uses low-overhead UDP datagrams, making it ideal for low-memory microcontrollers.
3. **AMQP (Advanced Message Queuing Protocol):** An open-standard, enterprise-grade messaging protocol providing queued, reliable, and secure messaging. Supports both publish-subscribe and point-to-point queues with message delivery acknowledgments, transaction tracking, and TLS/SASL security.
4. **HTTP / REST & WebSockets:** Standard web application protocols utilized in IoT platforms for web/mobile user integration. REST over HTTP provides familiar request-response APIs, while WebSockets offer full-duplex persistent connections for real-time dashboard updates.
5. **CBOR (Concise Binary Object Representation):** A binary data encoding format (RFC 7049) designed to provide JSON-like data structuring with vastly reduced payload size. Compresses text keys into binary representations, reducing parsing complexity and bandwidth consumption on constrained microcontrollers.
6. **JSON and XML Formats:** Human-readable text-based data representation formats widely used at the cloud and application layers. Provide high syntactic interoperability and easy debugging, though they impose higher payload size overhead compared to binary CBOR.
7. **OMA LwM2M (Lightweight Machine-to-Machine):** An application-layer device management protocol designed by the Open Mobile Alliance for constrained hardware. Built directly on top of CoAP and UDP, utilizing a client-server object model to handle remote provisioning, monitoring, and OTA updates.

8. **Management IDs in OMA LwM2M:** Unique logical resource identifiers assigned at the management tier. Used by LwM2M servers to authenticate endpoints, perform parameter configuration, read diagnostic metrics, and push firmware files.
9. **OPC UA (Unified Architecture):** An industrial M2M communication standard delivering secure, platform-independent data exchange across smart manufacturing systems. Provides semantic interoperability, structured domain data models, and built-in cryptographic security for Industrial IoT (IIoT).
10. **oneM2M Global Service Layer Standard:** A global initiative standardizing the common service layer architecture for M2M and IoT platforms. Defines reusable service functions (data management, device abstraction, security) across cellular and enterprise IoT networks.

Section B: 5-Mark Questions

Q6. Describe the sequential steps involved in building an end-to-end IoT Architecture.

Answer:



1. **Define Objectives & Use Case:** Clarify the specific real-world problem being solved (e.g., smart agriculture monitoring) to extract requirements like data sampling rates, network latency, and budget constraints.
2. **Identify Perception Devices:** Select hardware sensors and actuators based on physical environmental variables while evaluating hardware power, memory, and environmental ruggedness.
3. **Select Communication & Network Protocols:** Choose connectivity technologies (BLE, LoRaWAN, MQTT, CoAP) by balancing communication range, bandwidth, reliability, and energy constraints.
4. **Add Edge Layer Gateways:** Deploy edge hardware nodes between sensors and the cloud to handle local data aggregation, noise filtering, and offline operational logic.
5. **Design Middleware / Cloud Core:** Provision core cloud platforms (e.g., AWS IoT Core) to handle secure device registration, time-series storage pipelines, and predictive analytics.

Q7. Explain the concept of Interoperability in IoT and describe its four distinct functional levels.

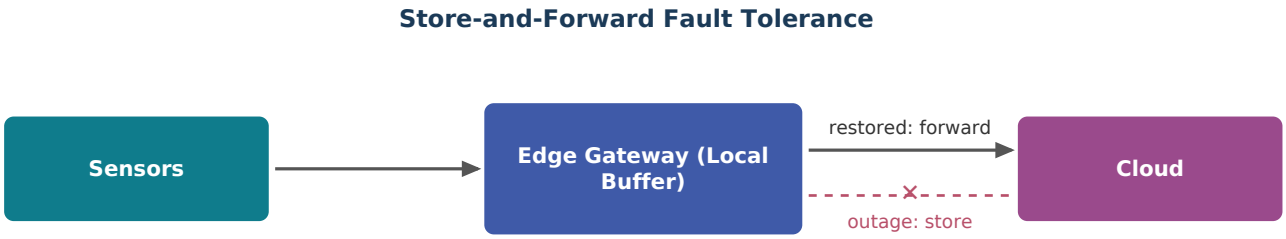
Answer:



1. **Technical Interoperability:** Focuses on hardware and lower-layer communication protocol compatibility, ensuring physical signals connect across technologies like Wi-Fi, BLE, and Zigbee.
2. **Syntactic Interoperability:** Defines standardized, predictable data formats and encodings (e.g., JSON, XML, CBOR) so receiving platforms parse message structures without errors.
3. **Semantic Interoperability:** Ensures the true conceptual meaning of transmitted data is preserved and understood across systems using shared vocabularies and data models.
4. **Organizational Interoperability:** Aligns business workflows, operational policies, and cross-organizational governance rules so distinct commercial entities collaborate via IoT insights.
5. **Importance in IoT Ecosystems:** High interoperability prevents proprietary vendor lock-in, reduces infrastructure costs, and allows seamless multi-vendor hardware integration.

Q8. How do IoT architectures achieve Reliability and Fault Tolerance in lossy network environments? Explain five key techniques.

Answer:



1. **Store-and-Forward Strategy:** Edge gateways buffer collected sensor telemetry locally during network outages and forward it automatically to the cloud once connectivity restores.
2. **Retries with Exponential Backoff:** Failed message transmissions trigger automatic retry sequences using exponentially increasing wait intervals to avoid flooding unstable networks.
3. **Edge Data Buffering:** Intermediate edge nodes smooth out irregular network transmission bursts and offload raw processing from cloud servers.
4. **Cloud Database & Server Redundancy:** Cloud platforms replicate device telemetry across geographically distributed data centers, ensuring seamless failover if one server goes down.

5. **Quality of Service (QoS) Guarantees:** Transport protocols enforce message delivery assurances (e.g., MQTT QoS levels 1 and 2) to guarantee critical commands are delivered at least once or exactly once.

Q9. Compare the LPWAN technologies LoRaWAN, NB-IoT, and LTE-M across technical parameters in a structured table format.

Answer:

Technical Parameter	LoRaWAN	NB-IoT	LTE-M (Cat-M1)
Spectrum Type	Unlicensed ISM Bands (868/915 MHz)	Licensed Cellular Spectrum (4G LTE)	Licensed Cellular Spectrum (4G LTE)
Data Rate	Low (0.3 kbps – 50 kbps)	Medium-Low (up to ~250 kbps)	High (up to ~1 Mbps)
Coverage Range	Extreme (10–15 km Rural, 2–5 km Urban)	High (Deep indoor & rural coverage)	Good (Slightly less indoor depth than NB-IoT)
Mobility & Voice	Static / Limited Mobility; No Voice	Static sensors only; No Voice	Full Cellular Mobility & VoLTE Voice support
Battery Lifetime	Ultra-High (10+ years)	High (10+ years)	High (10+ years)

1. **Spectrum Usage:** LoRaWAN operates on free unlicensed ISM bands, whereas NB-IoT and LTE-M rely on carrier-managed licensed cellular spectrum.
2. **Data Throughput:** LTE-M delivers the highest bandwidth (1 Mbps), making it suitable for asset tracking, while LoRaWAN handles lightweight periodic telemetry.
3. **Mobility Support:** LTE-M supports seamless cellular handover for moving assets, whereas NB-IoT and LoRaWAN target stationary nodes.
4. **Deployment Costs:** LoRaWAN allows private network setups without subscription fees, whereas cellular options incur SIM charges.
5. **Primary Application:** LoRaWAN excels in smart agriculture, NB-IoT in utility metering, and LTE-M in wearable healthcare and fleet management.

Q10. Compare Bluetooth Low Energy (BLE) and Classic Bluetooth across operational metrics in a structured table format.

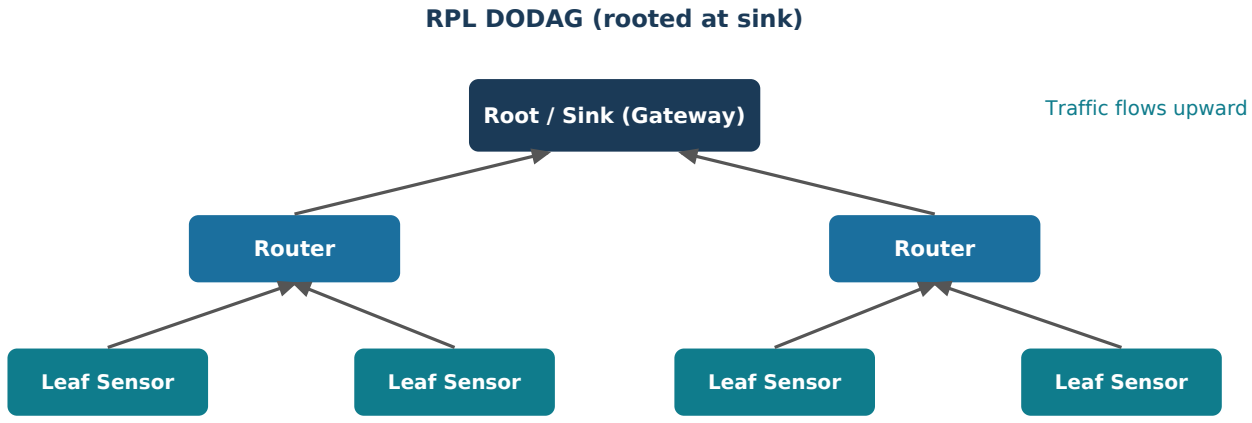
Answer:

Operational Metric	Bluetooth Low Energy (BLE)	Classic Bluetooth (BR/EDR)
Power Consumption	Ultra-Low (Runs years on coin cells)	High (Requires frequent battery recharging)
Data Throughput	~1 Mbps (Optimized for small bursts)	Up to 3 Mbps (Continuous data streams)
Connection Setup	Instantaneous (Fast wake up & sleep)	Slower connection setup time
Supported Topologies	Point-to-Point, Broadcast, Mesh	Point-to-Point, Piconet
Primary Target Use	Battery IoT sensors, wearables, beacons	Wireless audio streaming, file transfer

1. **Power Consumption:** BLE is engineered to sleep continuously and wake for rapid data bursts, conserving energy compared to Classic Bluetooth.
2. **Data Streaming:** Classic Bluetooth supports continuous high-bandwidth streams like audio, whereas BLE handles small bursty payloads.
3. **Network Topology:** BLE natively supports mesh topology for building automation, whereas Classic Bluetooth is restricted to point-to-point links.
4. **Wake-up Speed:** BLE connects and transmits in milliseconds, minimizing power drawer duration.
5. **Use Case Domain:** Classic Bluetooth powers headsets/speakers; BLE powers IoT smart beacons, medical monitors, and sensors.

Q11. Explain the architecture, features, and routing methodology of RPL (Routing Protocol for Low-Power and Lossy Networks).

Answer:

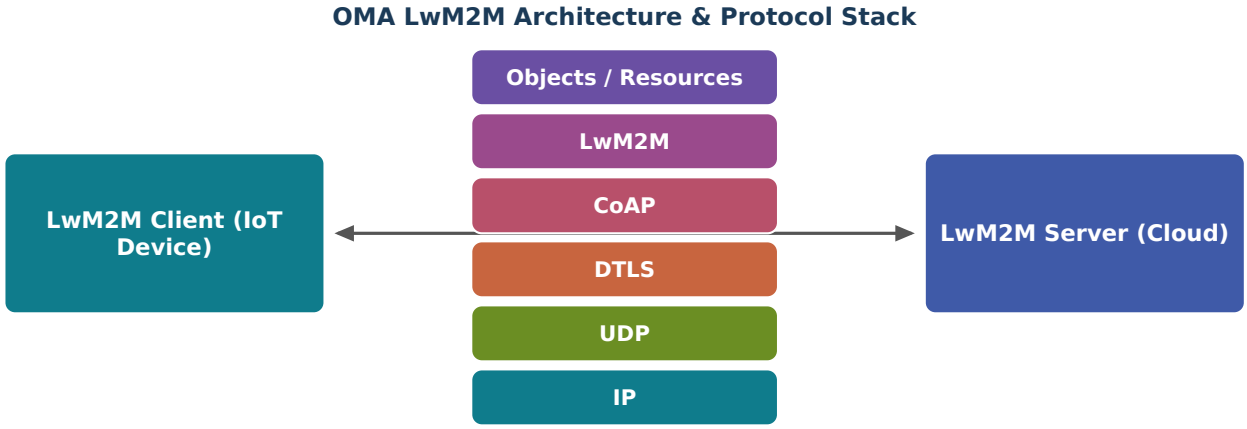


1. **IPv6 Integration via 6LoWPAN:** RPL operates natively with IPv6 over constrained 6LoWPAN links, allowing direct IP addressability for sensor nodes.
2. **DODAG Tree Structure:** RPL constructs Destination-Oriented Directed Acyclic Graphs (DODAGs) rooted at a central sink node to organize network traffic paths.
3. **Multipoint-to-Point Traffic Optimization:** Optimized primarily for gathering telemetry from many dispersed leaf sensors and routing it uphill to a single gateway.

4. **Objective Functions (OFs):** Selects optimal routing paths based on metrics such as link quality (ETX), hop count, and node remaining energy levels.
5. **Low Control Overhead:** Minimizes routing control message frequency during stable states to conserve battery power and limited wireless bandwidth.

Q12. Describe the framework and protocol stack of OMA LwM2M for constrained device management.

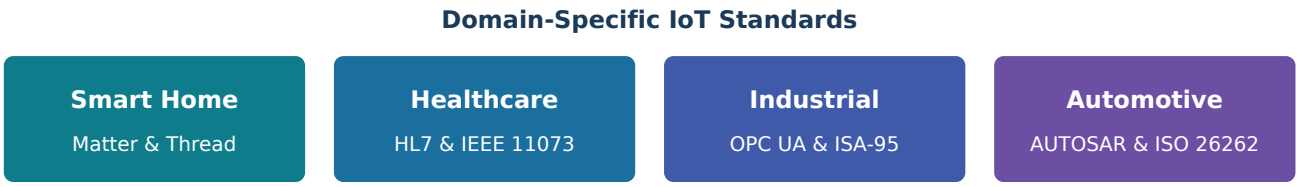
Answer:



1. **Client-Server Architecture:** Functions via an LwM2M Client running on the constrained IoT node communicating with a centralized LwM2M Cloud Server.
2. **Protocol Stack Dependencies:** Built directly on top of the CoAP application protocol, utilizing UDP datagrams over IP networks to minimize protocol overhead.
3. **Resource & Object Data Model:** Structures device features, sensor readings, and firmware attributes into standard, numbered logical Objects and Resources.
4. **Core Management Features:** Provides built-in functions for remote device onboarding, parameter configuration, battery status monitoring, and OTA firmware updates.
5. **DTLS Security Integration:** Secures transport layer control traffic using Datagram Transport Layer Security (DTLS) for mutual authentication and payload encryption.

Q13. Discuss domain-specific IoT standards across Smart Home, Healthcare, Industrial, and Automotive verticals.

Answer:



1. **Smart Home - Matter & Thread:** Unified IP-based application standards ensuring cross-vendor interoperability for smart lighting, locks, and appliances.
2. **Healthcare IoT - HL7 & IEEE 11073:** Standards governing personal health telemetry exchange and medical device communications to preserve data accuracy and patient privacy.
3. **Industrial IoT (IIoT) - OPC UA & ISA-95:** Frameworks delivering semantic M2M interoperability and enterprise-to-shop-floor control integration for automated manufacturing.
4. **Automotive IoT - AUTOSAR & ISO 26262:** Software architecture and functional safety standards ensuring ECU reliability and vehicular safety compliance in connected vehicles.
5. **Strategic Value of Industry Standards:** Adopting domain standards simplifies regulatory compliance, ensures operational safety, and avoids vendor lock-in.

Section C: 2-Mark Questions

Q14. What is the functional role of the Edge Layer in an IoT architecture?

Answer:



1. **Local Data Processing:** Aggregates, filters, and cleans raw sensor stream data locally to reduce bandwidth consumption before cloud upload.
2. **Low-Latency Actuation:** Executes time-sensitive control loops locally, enabling continuous offline operations during cloud network outages.

Q15. Define 'Security by Design' in IoT and name two implementation mechanisms.

Answer:

1. **Core Concept:** Embeds identity, encryption, and firmware protection directly into hardware and software components during initial design.
2. **Key Implementation Mechanisms:** Enforces cryptographically unique hardware device IDs and Secure Boot validation for firmware binaries.

Q16. What is CBOR and why is it preferred over JSON in constrained IoT devices?

Answer:

Concise Binary Object Representation (CBOR) is a compact binary data format defined in RFC 7049.

1. **Binary Efficiency:** Encodes structured data into small binary payloads, significantly reducing message sizes compared to text-based JSON.
2. **Resource Savings:** Minimizes parsing CPU cycles and transmission power consumption on memory-constrained microcontrollers.

Q17. Explain the function and operational frequency of Z-Wave in home automation.

Answer:

1. **Sub-GHz Operation:** Operates on sub-GHz ISM bands (868/908 MHz), eliminating signal interference from household Wi-Fi and BLE networks.
2. **Mesh Topology:** Connects smart home sensors and locks via reliable mesh routing using AES-128 encryption.

Q18. Define OMA LwM2M and specify its lower-layer protocol dependencies.

Answer:

1. **Definition:** Provides lightweight remote device provisioning, health monitoring, and firmware update capabilities.
2. **Protocol Stack:** Operates directly on top of CoAP over UDP and IP networking layers.

Q19. What is 6LoWPAN and what primary problem does it solve in IoT?

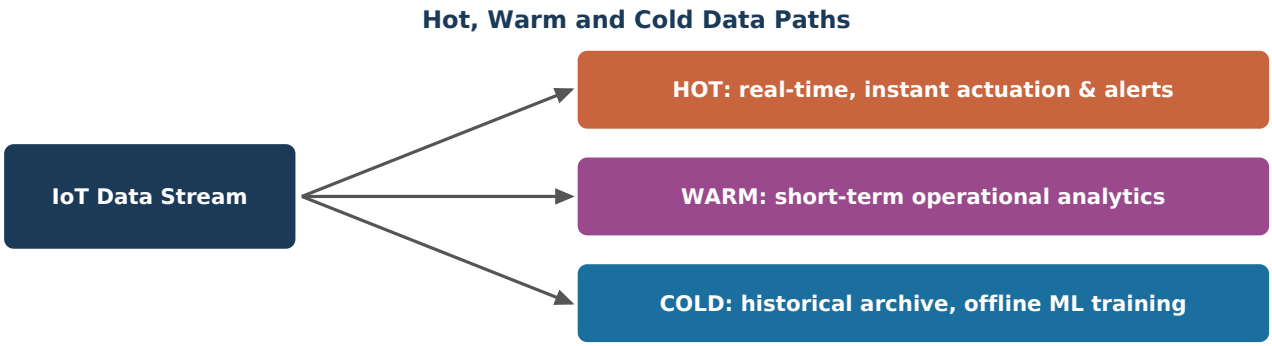
Answer:

6LoWPAN stands for IPv6 over Low-Power Wireless Personal Area Networks.

1. **Core Function:** Provides header compression and payload fragmentation protocols for IEEE 802.15.4 link layers.
2. **Problem Solved:** Allows tiny, low-power sensor nodes to receive direct, individual IPv6 web addresses.

Q20. Differentiate between Hot, Warm, and Cold data paths in IoT data-driven design.

Answer:



1. **Hot Data Path:** Processes real-time data streams immediately at the edge or cloud for instant control actuation and emergency alerts.
2. **Warm & Cold Data Paths:** Warm paths handle short-term operational analytics; Cold paths archive historical data in batch storage for offline ML model training.

Q21. Explain the purpose of the OpenIoT and FIWARE open-source middleware platforms.

Answer:

1. **OpenIoT Platform:** Implements a "Sensing-as-a-Service" paradigm to connect heterogeneous sensors and stream standardization.
2. **FIWARE Platform:** Delivers standardized Generic Enabler components for smart city data management and context modeling.

Q22. What is OPC UA and where is it primarily deployed?

Answer:

1. **Core Function:** Delivers platform-independent, cryptographically secured semantic data exchange across automation hardware.
2. **Primary Deployment:** Deployed extensively in Industrial IoT (IIoT), smart factories, and manufacturing control networks.